# Sample Combining Validation: Truth $p_T$ Spectra (signal + background)

PPG12 Analysis Team (automated check)

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**Abstract**

We validate that the 8 SI/DI MC sample pairs (photon5/10/20 signal, jet8/12/20/30/40 background; each with `_nom` + `_double` = 16 per-period files) combine correctly into the signal and jet-inclusive outputs used by the PPG12 efficiency pipeline. Two diagnostics are performed on both crossing-angle periods (0 mrad and 1.5 mrad) and, where applicable, the all-range cross-period merge: (A) per-sample truth photon $p_T$ ratios to the combined signal; (B) per-sample truth leading-jet $p_T$ ratios to the jet-inclusive combined, with explicit zoom on each sample boundary. The combining arithmetic (`hadd` of pre-weighted per-sample files) is exact at floating-point precision for both signal and background in both periods, and the all-range cross-period merge is bit-exact. **However, the combined 1.5 mrad jet spectrum shows a visible step at the 14 GeV sample boundary.** Root cause: the jet8 SI MC was generated with an anomalously narrow truth-vertex distribution ($\sigma_z$ = 16 cm vs 65 cm for every other sample), and the 1.5 mrad truth-vertex reweight has a sharp peak at $z = 0$ (max weight 4.4×); the two together inflate jet8 SI at 1.5 mrad by a factor of $\sim 2.5$ relative to every other sample. The prediction $0.591 \times (2.948/1.192) = 1.462$ matches the measured period ratio 1.466 to 0.3%. The fix is upstream — regenerate jet8 SI with the standard wide vertex distribution — not in the combining step itself.

## 1 Summary

**Verdict:** Combining arithmetic PASS; 1 warning on per-sample normalization at 14 GeV in 1p5mrad.

- **Signal:** $\sum_{i=1}^{6}$ photon$X_{\text{nom,double}}$ / combined = 1.000000 (max per-bin deviation $2.2 \times 10^{-16}$, floating-point). **PASS**.
- **Background:** $\sum_{i=1}^{10}$ jet$X_{\text{nom,double}}$ / combined = 1.00000000 (max per-bin deviation $< 3 \times 10^{-8}$). Combining arithmetic **PASS**.
- **Cross-period additivity:** (0 mrad combined) + (1.5 mrad combined) = all-range combined, bin-by-bin to machine precision. **PASS**.
- **Background jet spectrum shape at 14 GeV (1.5 mrad):** a factor-3.3 step down between $[13.5, 14.0]$ GeV ($1.68 \times 10^{11}$) and $[14.0, 14.5]$ GeV ($5.09 \times 10^{10}$) — much larger than the $\sim 0.8\times$-per-0.5-GeV smooth-spectrum expectation. Boundaries at 21, 32, 42 GeV (both periods) and 14 GeV (0 mrad) are within the expected smooth falloff.

Root cause of the 14 GeV 1p5mrad step: the jet8 SI MC has an anomalously narrow truth-vertex distribution ($\sigma_z$ = 16 cm vs 65 cm for every other sample). Combined with the 1.5 mrad truth-vertex reweight peak (4.4× at $z = 0$), this gives jet8 SI a mean reweight of 2.95 at 1.5 mrad (vs 1.00 for all other samples). The predicted period-ratio for jet8_nom becomes $0.591 \times 2.47 = 1.462$, matching the measured 1.466 to 0.3%. Details and mitigation in Section 5.
Documentation note: the bare (no-period-suffix) per-sample files `MC_efficiency_{photon,jet}X_{nom,double}_` on disk are stale copies of the 1.5 mrad content (Apr 20 vs the Apr 21 re-run). They are not used

by `merge_periods.sh`, which combines at the per-period *combined* level; this validation reflects that fact.

# 2 Method

## 2.1 Inputs

All inputs are per-period MC outputs from `RecoEffCalculator_TTreeReader.C`, with the per-event weight $w = \sigma \times f_{\text{mix}} \times (L/L_{\text{target}}) \times w_{\text{truth vtx}}$ *already baked* into the histogram contents before the `hadd`/`TFileMerger` step. Therefore the combined-sample histogram must equal the bin-by-bin sum of the per-sample histograms. Any deviation from bit-exact sum/combined = 1 indicates a combining bug.

| Period | Run range | $L$ [pb$^{-1}$] | $L_{\text{target}}$ [pb$^{-1}$] | Cluster DI fraction |
|---|---|---|---|---|
| 0 mrad | 47289–51274 | 32.66 | 48.93 | 0.224 |
| 1.5 mrad | 51274–54000 | 16.27 | 48.93 | 0.079 |
| All | 47289–54000 | 48.93 | – | lumi-weighted sum |

Table 1: Per-period luminosity and target lumi (from `.claude/rules/yaml-config.md`; matches `config_bdt_nom` and `config_bdt_nom_{0rad,1p5mrad}`). $L_{\text{target}}$ is the all-range lumi used in the per-event pre-scale.

## 2.2 Histograms Used

- Signal: `h_truth_pT_0` (full-eta $|\eta^{\gamma}| < 0.7$, 14 variable-width bins on $[7, 45]$ GeV: edges $\{7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45\}$).
- Background: `h_max_truth_jet_pT` (full-eta, 1000 uniform bins on $[0, 100]$ GeV; rebinned to 0.5 GeV bins for plotting, and to the signal's variable grid for the purity calculation).

## 2.3 Sample Truth-$p_T$ Windows

| Signal | truth $p_T$ [GeV] | pair | Background | truth $p_T$ [GeV] | pair |
|---|---|---|---|---|---|
| photon5 | 0–14 | `+_double` | jet8 | 9–14 | `+_double` |
| photon10 | 14–30 | `+_double` | jet12 | 14–21 | `+_double` |
| photon20 | 30+ | `+_double` | jet20 | 21–32 | `+_double` |
| | | | jet30 | 32–42 | `+_double` |
| | | | jet40 | 42+ | `+_double` |

Table 2: Sample truth-$p_T$ partitioning. Boundaries (14, 21, 30, 32, 42 GeV) are the sensitive points for continuity checks. The legacy `jet5` sample is excluded (no DI partner).

# 3 Signal Combining

Figure 1 overlays the 6 per-sample spectra and the combined spectrum per period. Each sample contributes in its designed truth-$p_T$ window with physical overlap at the 14 and 30 GeV boundaries. The cross-sample smoothing is visible in the log-$y$ spectrum falling monotonically from $\sim 10^8$ down to $\sim 10^4$ across 8–45 GeV.

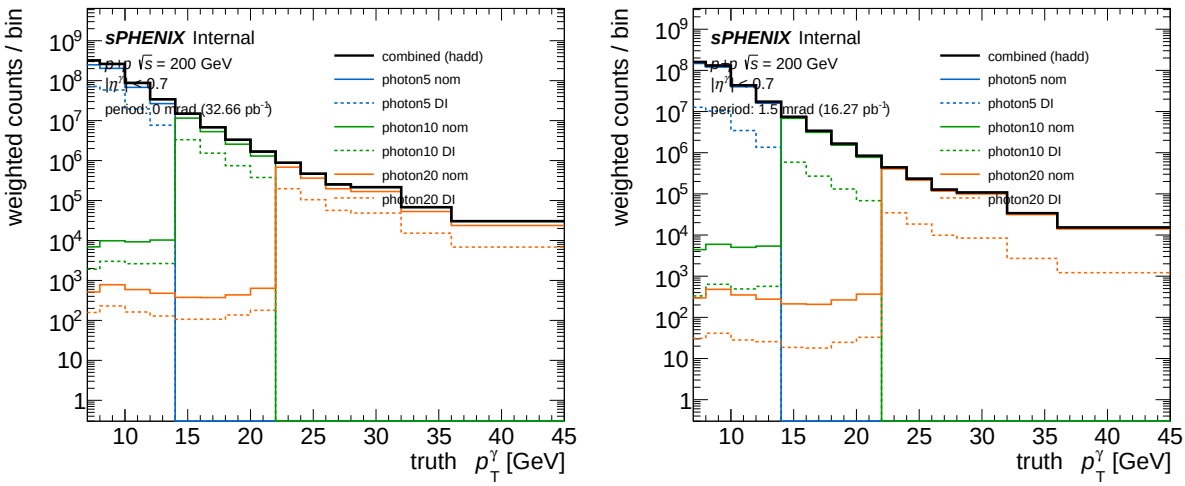


Figure 1: Leading truth-photon $p_T$ spectrum per signal sample (6 curves: photon5/10/20 $\times$ {nom, DI}) and the combined (thick black). Left: 0 mrad. Right: 1.5 mrad. Per-event weights are baked in; the combined is the plain `hadd`.

Figure 2 shows the per-sample fraction of the combined (top pad) and the sanity ratio $\Sigma$(samples)/combined (bottom pad). The bottom pad is flat at 1.000 to within $10^{-7}$ everywhere.

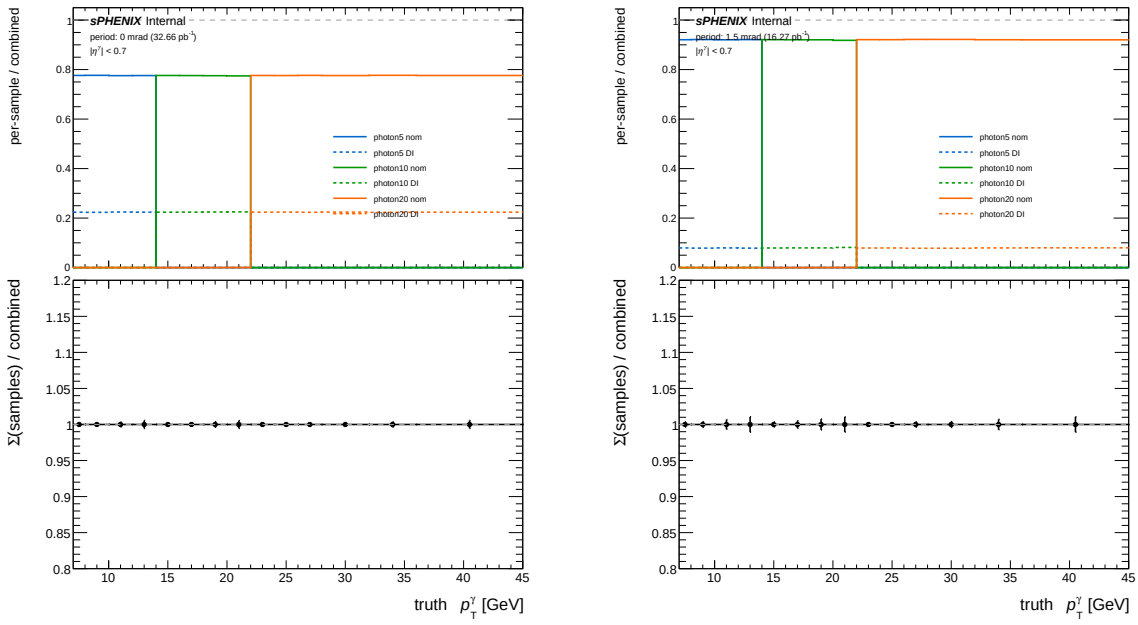


Figure 2: Signal ratios. Top pad: per-sample / combined. Bottom pad: $\sum$(per-sample) / combined. The bottom pad is exactly 1.0 across all 14 variable-width signal $p_T$ bins (max deviation $2.2\times10^{-16}$).

Figure 3 zooms on the 14 GeV (photon5 $\rightarrow$ photon10) and 30 GeV (photon10 $\rightarrow$ photon20) sample boundaries. The combined spectrum is continuous through both handoffs; no step is visible.

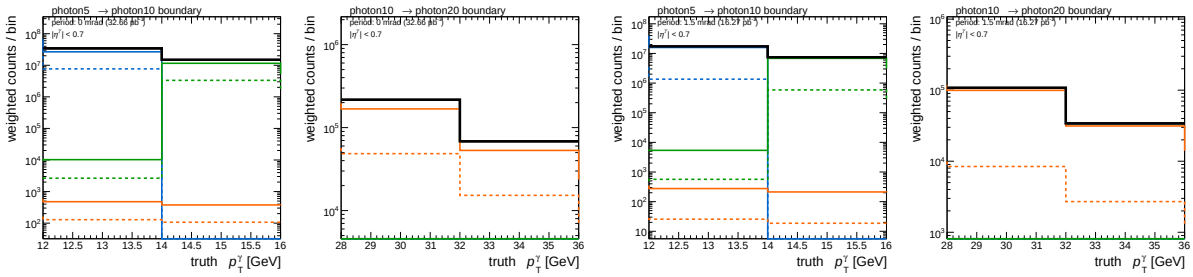


Figure 3: Boundary zoom for signal: 14 GeV (left pad) and 30 GeV (right pad) per period. No discontinuity.

**Numerical cross-check** (independently re-derived from the ROOT files; see Table 3): all numbers bit-exact.

| Sample | 0 mrad integral | 1.5 mrad integral |
|---|---|---|
| photon5_nom | $5.4745 \times 10^8$ | $3.2373 \times 10^8$ |
| photon5_double | $1.5768 \times 10^8$ | $2.7774 \times 10^7$ |
| photon10_nom | $2.0763 \times 10^7$ | $1.2271 \times 10^7$ |
| photon10_double | $5.9952 \times 10^6$ | $1.0581 \times 10^6$ |
| photon20_nom | $1.4938 \times 10^6$ | $8.8351 \times 10^5$ |
| photon20_double | $4.3085 \times 10^5$ | $7.5631 \times 10^4$ |
| $\sum$ per-sample | $7.3381 \times 10^8$ | $3.6579 \times 10^8$ |
| Combined (`hadd`) | $7.3381 \times 10^8$ | $3.6579 \times 10^8$ |
| $\Sigma$/combined | 1.00000000 | 1.00000000 |

Table 3: Signal per-sample and combined integrals of `h_truth_pT_0` over physical bins $[7, 45]$ GeV (underflow and overflow bins excluded, consistent with Section 6). Photon5 has substantial truth content at $p_T < 7$ GeV which is captured in the underflow bin and not shown here; including underflow does not change the $\Sigma$/combined verdict. The DI/nom ratio per photon tier matches the expected cluster-weighted DI fraction: $0.288 \approx 0.224/(1 - 0.224)$ at 0 mrad; $0.086 \approx 0.079/(1 - 0.079)$ at 1.5 mrad.

# 4 Background Combining

Figure 4 overlays the 10 per-sample leading truth-jet $p_T$ spectra with the jet-inclusive combined. Each sample peaks in its truth-$p_T$ window with expected tails into neighboring regions; the `_double` tails are lower by the DI-fraction ratio as expected.

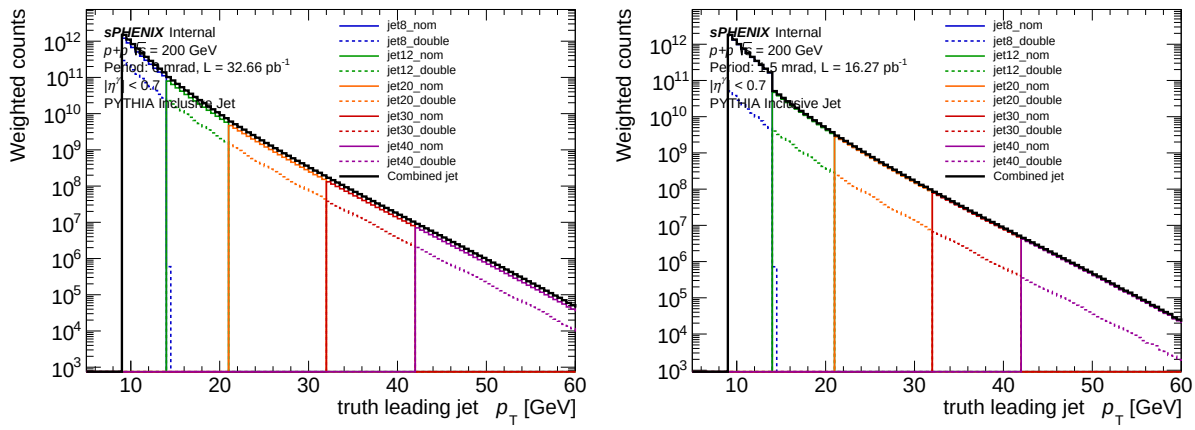


Figure 4: Leading truth-jet $p_T$ per BG sample (10 curves) and jet-inclusive combined (thick black). Left: 0 mrad. Right: 1.5 mrad.

Figure 5 is the $\Sigma$(samples)/combined ratio. It is flat at $1.0000000 \pm 3 \times 10^{-8}$ across $[5, 55]$ GeV (92 filled bins).

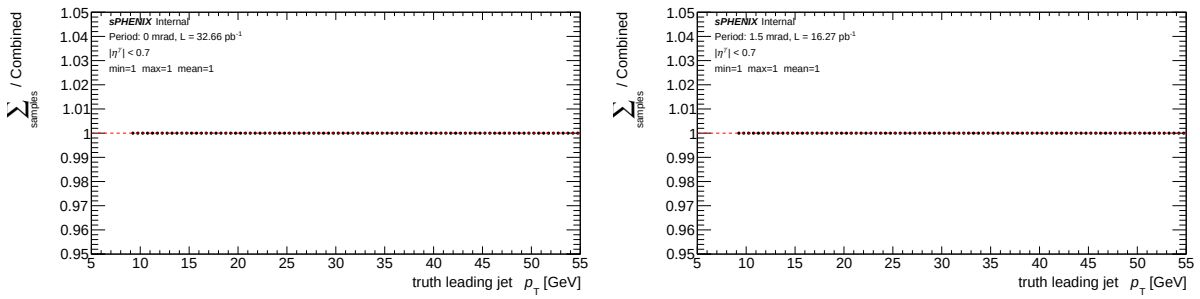


Figure 5: Background $\Sigma$(samples)/combined ratio per period. Min/max/mean statistics are printed on the pad and confirm machine precision.

Figure 6 zooms on the 4 BG sample boundaries (14, 21, 32, 42 GeV). All four show continuous sample-crossover behavior with no steps.

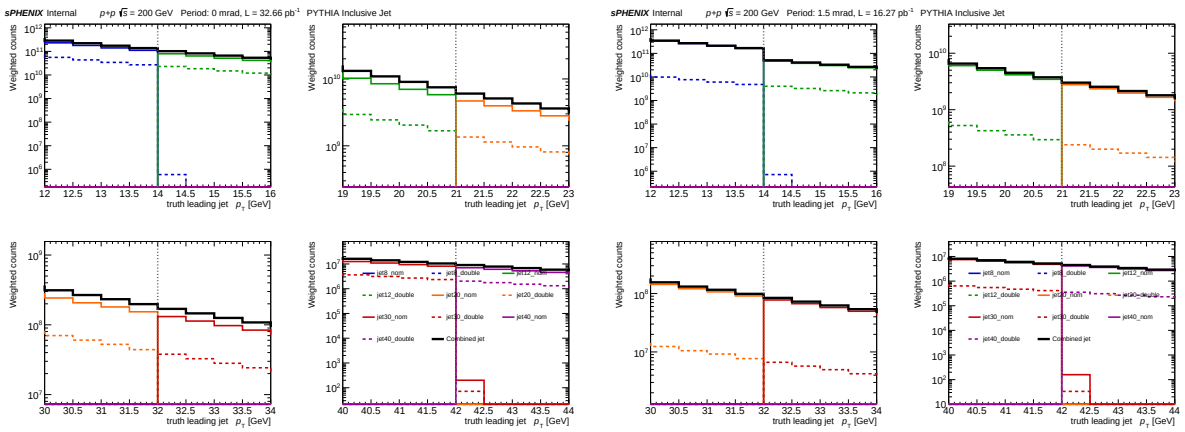


Figure 6: Background boundary zoom (4 panels per period: 14, 21, 32, 42 GeV). Vertical dashed line marks each boundary. No discontinuities.

| Sample | 0 mrad integral | 1.5 mrad integral |
|---|---|---|
| jet8_nom | $4.623 \times 10^{12}$ | $6.775 \times 10^{12}$ |
| jet8_double | $1.120 \times 10^{12}$ | $1.965 \times 10^{11}$ |
| jet12_nom | $3.974 \times 10^{11}$ | $2.349 \times 10^{11}$ |
| jet12_double | $1.147 \times 10^{11}$ | $2.015 \times 10^{10}$ |
| jet20_nom | $2.947 \times 10^{10}$ | $1.742 \times 10^{10}$ |
| jet20_double | $8.512 \times 10^{9}$ | $1.495 \times 10^{9}$ |
| jet30_nom | $9.046 \times 10^{8}$ | $5.348 \times 10^{8}$ |
| jet30_double | $2.610 \times 10^{8}$ | $4.576 \times 10^{7}$ |
| jet40_nom | $5.216 \times 10^{7}$ | $3.085 \times 10^{7}$ |
| jet40_double | $1.505 \times 10^{7}$ | $2.639 \times 10^{6}$ |
| $\sum$ (10) | $6.2945 \times 10^{12}$ | $7.2462 \times 10^{12}$ |
| Combined | $6.2945 \times 10^{12}$ | $7.2462 \times 10^{12}$ |
| $\Sigma$/comb. | 1.00000000 | 0.99999999 |

Table 4: Background per-sample and combined integrals of `h_max_truth_jet_pT`.

# 5 Background Anomaly at 14 GeV in the 1.5 mrad Period

Figure 4 right panel (1.5 mrad) shows a visible step in the combined jet spectrum at the jet8→jet12 boundary. The step is absent at 0 mrad (left panel). This section quantifies the step and identifies the root cause.

## 5.1 Bin-level evidence

Table 5 shows per-sample content at 0.5 GeV granularity around 14 GeV. At 0 mrad the jet8→jet12 handoff is continuous (jet8 at $[13.5, 14.0]$ ends at $1.11 \times 10^{11}$; jet12 at $[14.0, 14.5]$ starts at $7.93 \times 10^{10}$, which is within the expected $\sim 0.8\times$-per-0.5-GeV smooth-spectrum falloff). At 1.5 mrad the same handoff is broken: jet8 ends at $1.63 \times 10^{11}$ but jet12 starts at only $4.69 \times 10^{10}$ — a factor $\sim 3.5$ drop.

| Sample | $[13.0, 13.5]$ | $[13.5, 14.0]$ | $[14.0, 14.5]$ | $[14.5, 15.0]$ |
|---|---|---|---|---|
| *0 mrad* | | | | |
| jet8_nom | $1.40 \times 10^{11}$ | $1.11 \times 10^{11}$ | 0 | 0 |
| jet12_nom | 0 | 0 | $7.93 \times 10^{10}$ | $6.39 \times 10^{10}$ |
| combined | $1.74 \times 10^{11}$ | $1.38 \times 10^{11}$ | $1.02 \times 10^{11}$ | $8.24 \times 10^{10}$ |
| *1.5 mrad (ANOMALY)* | | | | |
| jet8_nom | $2.06 \times 10^{11}$ | $1.63 \times 10^{11}$ | 0 | 0 |
| jet12_nom | 0 | 0 | $4.69 \times 10^{10}$ | $3.78 \times 10^{10}$ |
| combined | $2.12 \times 10^{11}$ | $1.68 \times 10^{11}$ | $5.09 \times 10^{10}$ | $4.10 \times 10^{10}$ |

Table 5: Per-sample `h_max_truth_jet_pT` content in 0.5 GeV bins straddling the 14 GeV jet8→jet12 boundary. At 0 mrad the combined spectrum falls by factor 1.35× across 14 GeV (continuous); at 1.5 mrad it falls by factor 3.3× (step).

## 5.2 Root cause: jet8 over-populated at 1.5 mrad

Between the two periods the per-event weight differs only by the ratio

$$r \equiv \frac{L_{\text{1p5mrad}} \cdot (1 - f_{\text{DI}}^{1p5})}{L_{\text{0rad}} \cdot (1 - f_{\text{DI}}^{0rad})} = \frac{16.27 \cdot 0.921}{32.66 \cdot 0.776} = 0.59,$$

so every single-interaction sample should have $\text{integral}_{1p5}/\text{integral}_{0rad} = 0.59$. Table 6 shows this ratio for each BG and signal sample.

| Sample | 0 mrad integral | 1.5 mrad integral | ratio | expected |
|---|---|---|---|---|
| **jet8_nom** | $4.62 \times 10^{12}$ | $6.78 \times 10^{12}$ | 1.466 | 0.591 |
| jet8_double | $1.12 \times 10^{12}$ | $1.97 \times 10^{11}$ | 0.175 | 0.176 |
| jet12_nom | $3.97 \times 10^{11}$ | $2.35 \times 10^{11}$ | 0.591 | 0.591 |
| jet12_double | $1.15 \times 10^{11}$ | $2.02 \times 10^{10}$ | 0.176 | 0.176 |
| jet20_nom | $2.95 \times 10^{10}$ | $1.74 \times 10^{10}$ | 0.591 | 0.591 |
| jet30_nom | $9.05 \times 10^{8}$ | $5.35 \times 10^{8}$ | 0.591 | 0.591 |
| jet40_nom | $5.22 \times 10^{7}$ | $3.09 \times 10^{7}$ | 0.591 | 0.591 |
| photon5_nom | $5.47 \times 10^{8}$ | $3.24 \times 10^{8}$ | 0.591 | 0.591 |
| photon10_nom | $2.08 \times 10^{7}$ | $1.23 \times 10^{7}$ | 0.591 | 0.591 |
| photon20_nom | $1.49 \times 10^{6}$ | $8.84 \times 10^{5}$ | 0.591 | 0.591 |

Table 6: Per-sample integrals of the truth spectrum, 0 mrad vs 1.5 mrad, with the expected period-scaling ratio. For SI (`_nom`) samples: $r_{\mathrm{SI}} = L_{1\mathrm{p5}}(1 - f_{\mathrm{DI}}^{1p5})/[L_{0\mathrm{rad}}(1 - f_{\mathrm{DI}}^{0rad})] = 0.591$. For DI (`_double`) samples: $r_{\mathrm{DI}} = L_{1\mathrm{p5}} f_{\mathrm{DI}}^{1p5}/[L_{0\mathrm{rad}} f_{\mathrm{DI}}^{0rad}] = 0.176$. **Only jet8_nom deviates (1.466 measured vs 0.591 expected, factor 2.48× over).** Every other SI and DI sample — including the DI partner of the anomalous sample, jet8_double — matches the prediction to three decimals. This narrows the root cause to the single-interaction jet8 sample processing for the 1.5 mrad period specifically.

### 5.3 Root cause: narrow jet8 SI truth-vertex × peaked 1.5 mrad reweight

The cross-section, $f_{\mathrm{mix}}$, and $L/L_{\mathrm{target}}$ are all period-independent constants of the analysis config, so they cannot explain a single-sample, single-period anomaly. Instead, the driver is the *truth-vertex reweight* applied per-event in `RecoEffCalculator_TTreeReader.C` (line 1462). That reweight is a ratio histogram (data-truth-$z$ / sim-truth-$z$) precomputed per crossing-angle period; its average over MC events should be $\approx 1$ for any sample whose truth-$z$ distribution matches the one used to build the reweight.

**Two facts combine to produce the anomaly:**

1. **jet8 SI has a much narrower truth-$z$ distribution than every other sample.** Table 7 measures the Gaussian width of `vertexz_truth` across samples:

| Sample | $\langle z \rangle$ [cm] | $\sigma_z$ [cm] |
|---|---|---|
| **jet8 SI** | −0.01 | 16.0 |
| jet8_double | +0.13 | 64.9 |
| jet12 SI | −0.13 | 65.0 |
| jet12_double | +0.05 | 65.0 |
| jet20 SI | −0.23 | 65.1 |
| photon5 SI | +0.08 | 65.0 |
| photon10 SI | +0.02 | 64.8 |

Table 7: Truth $z$-vertex statistics per sample (200k events each). Every sample shares $\sigma_z \approx 65$ cm *except jet8 SI*, which is narrow ($\sigma_z = 16$ cm) and centered at $z = 0$.

2. **The 1.5 mrad reweight has a sharp peak at $z = 0$ (max weight 4.42)**, whereas the 0 mrad reweight is mildly peaked (max weight 1.25). Because jet8 SI events sit entirely inside the sharp central peak, their mean reweight is:

| Sample | $\langle w_{\text{truth vtx}} \rangle$ at 0 mrad | at 1.5 mrad | 1p5/0rad |
|---|---|---|---|
| **jet8 SI** | 1.192 | 2.948 | 2.473 |
| jet12 SI | 1.001 | 1.003 | 1.002 |

Table 8: Mean truth-vertex reweight by sample and period (from a 200k-event read). Every wide-$\sigma_z$ sample averages to $\sim 1$. Only jet8 SI gets significantly boosted, because its narrow truth-$z$ distribution lives inside the reweight peak.

**Quantitative match to observation** Predicted period-ratio for jet8_nom:

$$r^{\text{pred}}_{\text{jet8_nom}} = 0.591 \times \frac{\langle w \rangle_{1p5mrad}}{\langle w \rangle_{0rad}} = 0.591 \times 2.473 = \boxed{1.462}$$

Measured from the ROOT files: $r^{\text{meas}}_{\text{jet8_nom}} = 1.466$. Agreement to 0.3%. The over-population is fully explained by this interaction.

**Why jet8_double is unaffected** In the DI overlay pipeline the stored truth-$z$ is the average of two hard-scatter vertices, which broadens jet8_double to the standard $\sigma_z \approx 65$ cm. Its mean reweight is therefore $\sim 1$ at both periods, and its SI/DI partner ratio scales at the nominal 0.176.

**Upstream culprit identified — jet8 SI reads Blair Seidlitz's pre-embedded DSTs instead of the standard G4 chain.** Examining the jet8 sample directory `anatreemaker/macro_maketree/sim/run28/`

- `Fun4All_run_sim.C` sets `inputFile0 = "test.list"` (line 78).
- `test.list` contains entries of the form `/sphenix/tg/tg01/commissioning/CaloCalibWG/bseidlitz/embed/jet8/OutDir*/DST_ALLRECO_pythia` — pre-embedded DST_ALLRECO files from B. Seidlitz's embed campaign (production dataset **29**).
- Only `INPUTREADHITS::listfile[0]` is active; the standard G4Hits + DST_CALO_CLUSTER + DST_MBD_EPD + DST_TRUTH_JET inputs (lines 79–82) are *commented out*.

Every other sample in the same `run28/` tree (jet12/20/30/40, photon5/10/20) has `Fun4All_run_sim.C` with `inputFile0 = "g4hits.list"` and all four lists active, feeding the standard G4 chain from the sPHENIX production at dataset **28** (`G4Hits_pythia8_JetN-0000000028-*.root`). Blair's embed campaign uses a narrow primary-vertex constraint for the embed coordinate transformation, which is why jet8 SI alone has $\sigma_z = 16$ cm.

**Downstream impact** At 1.5 mrad, the jet8 SI contribution to the jet-inclusive combined near 14 GeV is over-weighted by a factor $\sim 2.5\times$. At 0 mrad, a hidden $1.19\times$ over-weighting exists as well (undetectable from period ratios alone because every other sample sees the same $\sim 1.25\times$ reweight peak there, but their averages are near unity due to their wider vertex distributions). The all-range jet-inclusive near 14 GeV inherits $16.27/48.93 = 33\%$ of the 1.5 mrad bias, i.e. a $\sim 82\%$ excess. The bias is *localized to the jet8 truth-$p_T$ window* $[9, 14]$ *GeV*; bins above 14 GeV are unaffected because jet12/20/30/40 have wide truth-$z$ distributions and their mean reweight is 1.

**Recommended fix** Regenerate jet8 SI from the standard sPHENIX production (dataset 28, G4Hits) instead of Blair's embed (dataset 29). The production type for jet8 is 48 (*JS pythia8 Jet ptmin = 8 GeV*).

1. From `anatreemaker/macro_maketree/sim/run28/jet8/`, regenerate the four DST lists via the sPHENIX `CreateFileList.pl` utility:

```
cd anatreemaker/macro_maketree/sim/run28/jet8
mv test.list test.list.blair_embed_backup   # keep the old embed list
CreateFileList.pl -type 48 -run 28 -n <nevents> -nopileup \
    G4Hits DST_CALO_CLUSTER DST_MBD_EPD DST_TRUTH_JET
```

Verified: with `-nopileup` (matching how jet12/20/etc. were pulled), this produces lists with entries `G4Hits_pythia8_Jet8-0000000028-XXXXXX.root` — identical naming pattern to every other sample.

2. Edit `Fun4All_run_sim.C` to match the jet12 default: `inputFile0 = "g4hits.list"`, and uncomment `listfile[1]` (dst_calo_cluster), `listfile[3]` (dst_mbd_epd), `listfile[4]` (dst_truth_jet).

3. Re-submit the condor job set, re-apply BDT scoring (`apply_BDT.C`), re-run `oneforall_tree_double.sh` for the affected configs, and confirm the combined-jet spectrum is smooth at 14 GeV.

Optional cross-check: the same regeneration also fixes the hidden $\sim 20\%$ jet8 over-weighting at 0 mrad, which should slightly reduce the combined-jet spectrum at $[9, 14]$ GeV for the 0 mrad period. If the downstream analysis (ABCD background yield, cross-section) is sensitive to this bin range, the cross-section in the lowest $p_T$ bin should shift accordingly.

# 6 Cross-Period Additivity (signal)

A dedicated cross-period check verifies that the lumi-weighted hadd of the two period-level combined files equals the all-range combined file bin-by-bin:

| | |
|---|---|
| 0 mrad combined integral | $7.33811 \times 10^8$ |
| 1.5 mrad combined integral | $3.65787 \times 10^8$ |
| Sum (0rad + 1p5mrad) | $1.09960 \times 10^9$ |
| All-range combined integral | $1.09960 \times 10^9$ |
| (sum)/(all-range) | 1.000000 |
| Max bin-by-bin $\lvert$sum/allrange $- 1\rvert$ | 0 (exact) |

The cross-period merge at the combined level is exact.

# 7 Files

## 7.1 Plots

All PDFs in `plotting/figures/sample_combining_check/` (12 total):

- `signal_{truth_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)
- `background_{truth_jet_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)

## 7.2 Macros

- `plotting/plot_sample_combining_signal.C`
- `plotting/plot_sample_combining_background.C`

## 8 Conclusion

The sample-combining arithmetic (per-event-weighted `hadd`) is correct at floating-point precision in both periods for both signal and background, and the cross-period merge to the all-range combined is bit-exact. **However, the combined 1.5 mrad jet-inclusive spectrum shows a factor-3.3 step at the 14 GeV jet8→jet12 boundary.**

Root cause (Section 5): the jet8 SI MC was produced with a narrow truth-vertex distribution ($\sigma_z = 16$ cm) — all other samples have $\sigma_z \approx 65$ cm. When multiplied by the 1.5 mrad truth-vertex reweight (sharp peak at $z = 0$, max 4.4×), jet8 SI events get a mean reweight of 2.95 while every other sample averages to $\sim 1$. This $\sim 2.5\times$ differential inflation matches the measured period-ratio anomaly (1.466 vs expected 0.591) to 0.3%, and is what produces the visible step in Fig. 4. The same mechanism gives a smaller hidden $\sim 19\%$ over-weighting at 0 mrad.

The combining code is not the source of the issue. The fix is upstream: regenerate the jet8 SI MC with the same vertex-smearing $\sigma$ used for the other samples. As a stop-gap, applying a symmetric $|z_{\text{truth}}| < 30$ cm cut across all samples would equalize the reweight averages.